The Focus

How the threads of 27 October 2026 meet: one map, with the status of every link (placeholder title)



DICTO · "everything is the same study now: all aspects of everything we have been looking at have come to a focus" · inside the Quanta of Space frame (Atoms / Bonds / Shells / Cycles)

> **The one piece.** Everything below is built from the octet pair: the regular tetrahedron and the regular octahedron. A tetrahedron is an octahedron with four small tetrahedra on alternate faces; a regular octahedron sits inside every tetrahedron, at the midpoints of its edges. The rule of halves shares space into such pieces, whole or halved; the 600-cell is 600 tetrahedra, and each hides one octahedron.

Thread	Link to the focus	Status
Atoms — the octet pair	tetrahedron = octahedron + 4 corner tetrahedra; space filled by both (octet)	**classical**
Bonds — Rule of Halves (#57)	every stacking of balls shares space into pieces borrowed whole or in half from the cubic and hexagonal stackings	**proved**
Pyrochlore Halves (#60)	corner-sharing tetrahedra, 2 per corner; the rule of halves holds on it in every stacking	**proved**
Swap Rule, Ganesha (#61)	squashed stackings: mixed pieces swap head and body; mixed = pure in sum; 4 : 5 : 6 at true close packing	**proved**
Shells — 600-cell slices (#63)	62 slice patterns, exact; 12 all triangles; Humpty Dumpty, the 48-vertex egg	**proved**
Squares = octahedron equators	the 46 true squares in the slices are exactly the equators of the cells' inner octahedra	**proved (exact)**
600 octahedra, corner-sharing	the inner octahedra meet 5 per corner round 120 icosahedral holes (the rectified 600-cell, known): pyrochlore's pattern with octahedra	**known shape**
Shadows (#62, #63)	each shadow is a solid ball tiled twice: front cells once, back cells once (halves of the 600-cell)	**proved (exact)**
The Cleo shells (#63)	round one vertex the 600 cells form 15 shells, each one kind of piece, round a core of 20 Cleo-Icos filling an icosahedron; the shadow's front layer = core + six shells	**proved (exact)**
Five pyramids → Cleo core	the icosahedron's face pyramid (Cleo-Icos), the odd one out in flat space (#44), is the core piece of the 600-cell's shells; Aristotle's "twenty tetrahedra round a point" (Bradwardine) closes in 4D	**proved (exact) + history**
DICTO's radial cell projection	RCP builds the 600-cell cell by cell, folding at the angle with cosine $(1 + 3\sqrt{5})/8$ = the swap moment on the vertex–cell turning; its 16 steps from a cell = the adjacency shells of the exact model	**proved (exact)**
Six turnings	the six pairs of main views; 14 swap moments, 20 patterns between: every view accounted for; cells swap front/back in pairs — swaps under pressure	**proved (exact)**
Halves → Shadow-17 (#62)	cut in halves, each half a J11, the piece DICTO's work began with	**proved**
Kyushindo Cores (#43)	golden nested icosahedral shells; echoed by the 600-cell's slices and shadows	**to test**
Cycles — Gnomon Ladders	every turning angle has cos = $\sqrt{(a + b\sqrt{5})}$: buildable from gnomons; 18° = half the golden gnomon	**stage 0**
Fold angle	a swap on the vertex–cell turning sits exactly at the 600-cell's fold angle (180° − dihedral)	**proved (exact)**
Scutoids	Janus and Ganesha pieces are scutoids, as are pure pieces: a family likeness	**likeness only**
Pressure Swap	can swapped halves or swapped edges make the five impossible slices possible?	**queued**

One sentence

Space, in flat stackings and in the four-dimensional 600-cell alike, is shared out into a few kinds of pieces built from the tetrahedron and the octahedron, whole or in halves, and when a view or a stacking changes, the pieces change by swapping, never by appearing from nothing.

Status words: *proved* = exact proof in a published paper or committed study; *exact* = exact computation, not yet in a paper; *decimal* = floating-point check only; *to test*, *queued*, *stage 0* = planned. The closing sentence is a summary of the proved rows and an aim for the others, not a new claim.